

Side-by-side

Paul Kiesel
4/29/2019

WIDIA
STREAM

DIAWI
STREAM

Z O B K R U O
X O G H U L B
S O L I F B B
W F L R V Q Q
P R N G K S S
O T W T Q S J
Q S S E K Z Z
W A T J K L U
W I D I A N F
B N Y P V T T
M Z F P K W G
D K Z X T J C
D I G K U H U
A U E K C A R

Z O B K R U O
X O G H U L B
S O L I F B B
W F L R V Q Q
P R N G K S S
O T W T Q S J
Q S S E K Z Z
W A T J K L U
D I A W I N F
B N Y P V T T
M Z F P K W G
D K Z X T J C
D I G K U H U
A U E K C A R

A

Clues K1, K2 and K3 PTs revealed to insert S into K2 CT to get XLAYERTWO. Analysis of K2 PT and last 120 characters of K2 CT via shifts of ABSCISSA revealed that Z had to be inserted into the beginning of K4 CT. Note that K4 now has 98 characters.

I also determined via the relationship of DYHR with HYDRA that the string DIAWI in the modified K4 CT had to undergo a transposition resulting in WIDIA.

I determined later that both the WIDIA stream and the DIAWI stream are needed to go forward.

10 SHIFT

J Y L U B E Y
H Y Q R E V L
C Y V S P L L
G P V B F A A
Z B X Q U C C
Y D G D A C T
A C C O U J J
G K D T U V E
G S N S K X P
L X I Z F D D
W J P Z U G Q
N U J H D T M
N S Q U E R E
K E O U M K B

J Y L U B E Y
H Y Q R E V L
C Y V S P L L
G P V B F A A
Z B X Q U C C
Y D G D A C T
A C C O U J J
G K D T U V E
N S K G S X P
L X I Z F D D
W J P Z U G Q
N U J H D T M
N S Q U E R E
K E O U M K B

B

Clues from discovery of completion of Hoover's name around the periphery and proper application of the meaning of the K1 PT led to the conclusion that the Caesar shift of 10 version of K4 was the version of interest.

Also developed early on were probable keywords
THEBERLINCLOCK,
ABSCISS and
MENGLENHREUHR

MNSQ converts to MASK
via clues in K1 (AELSN, QELSK.
Also see MASP conv. to MASK
all in column 1 of 7-column pad).

KEY ABSCISS

C Q E F N O V
A Q H M V J E
Y Q J W H E E
S U J Z W R R
M K M E S P P
U Y A R M P X
K R P V S D D
S M T U S J O
S X G W E M W
D J C J W T T
J B W J S A H
F G D O U X F
F X H F V U O
N P Z F P Q Y

C Q E F N O V
A Q H M V J E
Y Q J W H E E
S U J Z W R R
M K M E S P P
U Y A R M P X
K R P V S D D
S M T U S J O
F X Q T L M W
D J C J W T T
J B W J S A H
F G D O U X F
F X H F V U O
N P Z F P Q Y

C

Doing a Quagmire III substitution with the above JYLUB... version and keyword ABSCISS, I arrived at this. (Note: The ABSCISS keyword was provided by a member of the Kryptos Group who asked that his name not be revealed. I thank him.)

NOTE: Matrix C is used as the CT in the OTP substitution coming on the next page.

WIDIA
STREAM

DIAWI
STREAM

KEY		QOQ				
I	K	L	M	Z	E	Y
G	K	Q	X	Y	V	L
B	K	V	P	Q	L	L
F	R	V	O	P	A	A
X	S	X	L	F	C	C
R	B	G	A	X	C	T
S	A	C	Y	F	J	J
F	X	D	R	F	V	E
F	T	N	P	L	X	P
J	V	I	V	P	D	D
V	H	P	V	F	G	Q
M	N	J	E	R	T	M
M	T	Q	M	Y	R	E
Z	C	O	M	C	K	B

D

I	K	L	M	Z	E	Y
G	K	Q	X	Y	V	L
B	K	V	P	Q	L	L
F	R	V	O	P	A	A
X	S	X	L	F	C	C
R	B	G	A	X	C	T
S	A	C	Y	F	J	J
F	X	D	R	F	V	E
M	T	K	D	W	X	P
J	V	I	V	P	D	D
V	H	P	V	F	G	Q
M	N	J	E	R	T	M
M	T	Q	M	Y	R	E
Z	C	O	M	C	K	B

This matrix has three columns that are the same as in matrix B. It serves as a clue to use matrix B.

K	E	O	U	M	K	B
N	S	Q	U	E	R	E
N	U	J	H	D	T	M
W	J	P	Z	U	G	Q
L	X	I	Z	F	D	D
G	S	N	S	K	X	P
G	K	D	T	U	V	E
A	C	C	O	U	J	J
Y	D	G	D	A	C	T
Z	B	X	Q	U	C	C
G	P	V	B	F	A	A
C	Y	V	S	P	L	L
H	Y	Q	R	E	V	L
J	Y	L	U	B	E	Y

E

K	E	O	U	M	K	B
N	S	Q	U	E	R	E
N	U	J	H	D	T	M
W	J	P	Z	U	G	Q
L	X	I	Z	F	D	D
N	S	K	G	S	X	P
G	K	D	T	U	V	E
A	C	C	O	U	J	J
Y	D	G	D	A	C	T
Z	B	X	Q	U	C	C
G	P	V	B	F	A	A
C	Y	V	S	P	L	L
H	Y	Q	R	E	V	L
J	Y	L	U	B	E	Y

Keyword of OTP using matrix C as CT

Flipped matrix B due to ZUG clue and VEER clue in matrix C and also VIRTUALLY in morse code slab.

OTP using Matrix C as CT and matrix E as keyword.

C	C	S	L	R	O	H
H	H	Q	W	E	I	K
C	Z	K	C	T	A	N
C	O	G	K	Y	H	A
R	Y	P	F	Q	N	N
B	V	H	U	M	O	U
G	R	N	M	E	H	Z
Z	C	U	J	E	K	I
T	H	K	G	T	C	N
E	B	E	V	Y	U	U
P	O	R	B	Q	K	A
P	E	H	Z	M	A	U
X	V	Q	E	E	Z	H
P	R	B	L	U	C	K

F

C	C	S	L	R	O	H
H	H	Q	W	E	I	K
C	Z	K	C	T	A	N
C	O	G	K	Y	H	A
R	Y	P	F	Q	N	N
Y	V	A	H	F	O	U
G	R	N	M	E	H	Z
Z	C	U	J	E	K	I
D	H	A	Q	D	C	N
E	B	E	V	Y	U	U
P	O	R	B	Q	K	A
P	E	H	Z	M	A	U
X	V	Q	E	E	Z	H
P	R	B	L	U	C	K

Result of OTP using above matrices

WIDIA STREAM

P X P P E T Z G B R C C H C
R V E O B H C R V Y O Z H C
B Q H R E K U N H P G K Q S
L E Z B V G J M U F K C W L
U E M Q Y T E E M Q Y T E R
C Z A K U C K H O N H A I O
K H U A U N I Z U N A N K H

G

DIAWI STREAM

P X P P E D Z G Y R C C H C
R V E O B H C R V Y O Z H C
B Q H R E A U N A P G K Q S
L E Z B V Q J M H F K C W L
U E M Q Y D E E F Q Y T E R
C Z A K U C K H O N H A I O
K H U A U N I Z U N A N K H

Rotated 90 degrees. According to the precedent set by matrices B and D, this matrix should have been chosen to be the CT of the OTP shown on Page 4, but it wasn't. See matrix L on next page to find out what happened.

KEY CLOCKTHEBERLIN

Q A X Q E K E Y K J B M Z J
M O S V B D U J H L T B Z J
Z P C M E V A B S M F C O G
B Q Q Z V C Y A G R Z M B X
F Q G E Y K W K D C R G V B
K B Y L U O F P W B G J K F
L W J X U I R H G B S Y E U

H

Q A X Q E S E Y Q J B M Z J
M O S V B D U J H L T B Z J
Z P C M E P A B Z M F C O G
B Q Q Z V J Y A S R Z M B X
F Q G E Y S W K T C R G V B
K B Y L U O F P W B G J K F
L W J X U I R H G B S Y E U

I already determined that CLOCKTHE-BERLIN was a keyword. Not having another clue, I tried it. It worked.

KEY AX

R O O M M X M K A H I J S H
Z P G Q I B Y H U I E S S H
S R J J M Q H S G J N A F E
I M R W P A C O Q Z S J I V
N M Q C C X T X L A B E P S
A S C I Y P N R T S Q H A D
X U W V Y G B F Q S G K M N

I

R O O M M T M K R H I J S H
Z P G Q I B Y H U I E S S H
S R J J M R H S S J N A F E
I M R W P H C O G Z S J I V
N M Q C C T T X E A B E P S
A S C I Y P N R T S Q H A D
X U W V Y G B F Q S G K M N

KEY ENLEHREUHRMENG

F X V P S Z P U U I A R Z R
D V T O P C G C C J P L Z R
L Q A R S U Z R R L E M O X
K E M B L B Q K B K R R B C
T E E Q W Z I N O B K V V N
M Z K K J T T V M A F Z K W
C H H A J H N A B A O E E S

J

F X V P S O P U I I A R Z R
D V T O P C G C C J P L Z R
L Q A R S Y Z R Q L E M O X
K E M B L I Q K R K X R B C
T E E Q W O I N Z B K V V N
M Z K K J T T V M A F Z K W
C H H A J H N A B A O E E S

Because of the TQSQSS anomaly in K4, I determined that MENGENLEH-REUHR was a probable keyword to be used as demonstrated.

Notice APEX and Zenith hints.

Note TREMBLING HANDS hint.

"The last part is the APEX of the Pyramid." Notice OBKR connected to APEX.

WIDIA STREAM

DIAWI STREAM

KEY Z

E	W	U	Y	O	X	Y	Q	Q	H	S
K	X	K	C	U	P	T	Y	B	F	B
B	I	Y	J	X	K	J	N	S	K	O
Q	X	K	K	J	D	L	T	W	Z	D
L	A	J	A	N	Z	A	Z	W	K	A
B	P	D	D	N	V	X	H	M	T	A
Z	U	U	M	L	X	Z	Z	I	P	P
U	L	S	E	X	Z	V		B	G	G
S	I	G	M	S	A	S	T	D	D	O

K

S	F	X	V	P	S	Z	P	U	U	I
A	R	Z	R	D	V	T	O	P	C	G
C	C	J	P	L	Z	R	L	Q	A	R
S	U	Z	R	R	L	E	M	O	X	K
E	M	B	L	B	Q	K	B	K	X	R
B	C	T	E	E	Q	W	Z	I	N	O
B	K	V	V	N	M	Z	K	K	J	T
T	V	M	A	F	Z	K	W	C	H	H
A	J	H	N	A	B	A	O	E	E	S

L

OTP

S	C	C	S	L	R	O	H	H	H	Q
W	E	I	K	C	Z	K	C	T	A	N
C	O	G	K	Y	H	A	R	Y	P	F
Q	N	N	B	V	H	U	M	O	U	G
R	N	M	E	H	Z	Z	C	U	J	E
K	I	T	H	K	G	T	C	N	E	B
E	V	Y	U	U	P	O	R	B	Q	K
A	P	E	H	Z	M	A	U	X	V	Q
E	E	Z	H	P	R	B	L	U	C	K

M

K	P	I	J	F	O	Q	I	A	A	U
D	J	D	R	R	W	T	V	Z	Y	Q
K	T	P	P	I	E	Q	J	M	T	I
F	Y	S	N	O	P	J	K	K	P	G
D	Z	J	S	Q	U	R	Z	O	B	J
B	Q	K	W	E	A	N	J	V	B	W
W	T	Q	R	X	I	Q	Z	M	V	T
W	N	A	N	G	A	N	Y	E	M	Q
V	O	I	O	T	A	Z	H	J	Y	S

N

E	W	U	Y	O	T	Y	Q	H	H	S
K	X	K	C	U	P	T	Y	B	F	B
B	I	Y	J	X	K	J	N	S	K	O
R	X	K	N	J	D	L	T	W	Z	D
L	A	J	H	N	Z	K	Z	W	K	A
B	P	D	D	N	V	T	H	M	X	A
Z	U	U	M	L	X	Z	Z	I	P	P
U	L	S	E	X	Z	V		B	G	G
S	I	G	M	S	A	S	T	D	D	O

S	F	X	V	P	S	O	P	U	I	I
A	R	Z	R	D	V	T	O	P	C	G
C	C	J	P	L	Z	R	L	Q	A	R
S	Y	Z	R	Q	L	E	M	O	X	K
E	M	B	L	I	Q	K	R	K	X	R
B	C	T	E	E	Q	W	O	I	N	Z
B	K	V	V	N	M	Z	K	K	J	T
T	V	M	A	F	Z	K	W	C	H	H
A	J	H	N	A	B	A	O	E	E	S

Keyword Z from Zenith hints in matrix J.

Says to add (SIGMA) S and S to beginnings (BGG) of the next two matrices

S added to beginning of matrix. Notice now 99 characters

This matrix is CT of OTP

S added to beginning of matrix. Notice now 99 characters.

This matrix is keyword of OTP

Another Quagmire III OTP substitution using SFXVP matrix as CT and SCCSL matrix as Keyword.

Note: If you repad this to nine columns, JUPITER, IO, GANYMEDE, EUROPA and JOVIAN appear in the matrix in various forms of readability,

some easy to read and others with the letters in groups. Analysis of this seemed to not lead anywhere. However, it is worth noting that all these are part of the JOVIAN System and IO, GANYMEDE and EUROPA are three of the largest moons of Jupiter. One other of the four largest moons is Callisto. It does not appear here.

Step N amplification: The following matrix is from the WIDIA stream. The DIAWI stream does not have all the Qs delimiting JUPITER. Also, the U is missing from the JUPITER cluster in the DIAWI stream.

K P I J F O Q I A	K P I J F O Q I A	K P I J F O Q I A
A U D J D R R W T	A U D J D R R W T	A U D J D R R W T
V Z Y Q K T P P I	V Z Y Q K T P P I	V Z Y Q K T P P I
E Q J M T I F Y S	E Q J M T I F Y S	E Q J M T I F Y S
N O P J K K P G D	N O P J K K P G D	N O P J K K P G D
Z J S Q U R Z O B	Z J S Q U R Z O B	Z J S Q U R Z O B
J B Q K W E A N J	J B Q K W E A N J	J B Q K W E A N J
V B W W T Q R X I	V B W W T Q R X I	V B W W T Q R X I
Q Z M V T W N A N	Q Z M V T W N A N	Q Z M V T W N A N
G A N Y E M Q V O	G A N Y E M Q V O	G A N Y E M Q V O
I O T A Z H J Y S	I O T A Z H J Y S	I O T A Z H J Y S
GANYMEDE AND IO	JUPITER AND JOVIAN	EUROPA AND IOTA

JUPITER AND MOONS.. ONLY IO IS SPELLED WITH LETTERS IN PROPER ORDER. VJYQTM IS ALSO SUSPECT.

K P I J F O Q I A	K P I J F O Q I A
A U D J D R R W T	A U D J D R R W T
V Z Y Q K T P P I	V Z Y Q K T P P I
E Q J M T I F Y S	E Q J M T I F Y S
N O P J K K P G D	N O P J K K P G D
Z J S Q U R Z O B	Z J S Q U R Z O B
J B Q K W E A N J	J B Q K W E A N J
V B W W T Q R X I	V B W W T Q R X I
Q Z M V T W N A N	Q Z M V T W N A N
G A N Y E M Q V O	G A N Y E M Q V O
I O T A Z H J Y S	I O T A Z H J Y S
SYSTEM (MAYBE)	VJYQTM (FROM K1)

I O T A

"And so, the further along we go, and the more the layers of the onion are unraveled, and the closer we get to cracking the Kryptos sculpture, the TINIER THE GRAIN is that would be responsible for cracking that code."

- Jim Sanborn in CNN 2005 interview

Q E M U F P H Z
L R F A X Y U
S D J K Z L D
K R N S H G N
F I V J Y Q T
Q U X Q B Q V
Y U V L L T R
E V J Y Q T M
K Y R D M F D
W CORRECT
P INCORRECT
K CORRECT

B E T W E E N
S U B T L E S
H A D I N G A
N D T H E A B
S E N C E O F
L I G H T L I
E S T H E N U
A N C E O F I
Q L U S I O N
L CORRECT
Q INCORRECT
L CORRECT

P A L I M P S
E S T P A L I
M P S E S T P
A L I M P S E
S T P A L I M
P S E S T P A
L I M P S E S
T P A L I M P
S E S T P A L
S CORRECT
C INCORRECT
C INCORRECT

E M U F P H Z	V J Y Q T Q	V - 4	GANYEM - YM
L R F A X Y U	N C E O F L	J - 8	JOVIAN - JV
S D J K Z L D	P A L I M P	Y - 3	JUPITER - JT
K R N S H G N		Q - 7	EUROPA - 0
F I V J Y Q T	V J Y Q T M	T - 6	IO - 0
Q U X Q B Q V	N C E O F I	M - 3	
Y U V L L T R	P A L I M P		
E V J Y Q T M			
K Y R D M F D			

VJYQTM - 3 TIMES

STUFF TO CONSIDER IN DECIDING WHAT TO DO NEXT

MISSPELLED LETTERS IN PYRAMID
MARKED BY Q IN ROW 6 OF 7-COL
PAD

		PYRAMID		PYRAMID	
	VJYQT			VJYQTM	
	FIVE				
EMUFPHZLRFAXYUSDJKZLDKRN	SHGNFIVJYQT	OUXQBOVYU	VLLTRE	VJYQTM	KYRDMF
BETWEENSUBTLESHADINGANDTHE	ABSENCE	OF	LIGHTL	IE	STHENUANCE
BETWEENSUBTLESHADINGANDTHE	ABSENTE	OF	LIGHTL	IE	STHENUANT
EESFIFTYSEVENMINUTESSIXPOINT	FIVE	SECONDS	NORTH	SEVENTYSEVEN	DEGREES

NUANT MEANS TWO THINGS TIED TOGETHER. CLUE QJNCE AT END OF ROW 1 OF K2 CT.

WIDIA STREAM

K	P	I	J	F	O	Q	I	A
A	U	D	J	D	R	R	W	T
V	Z	Y	Q	K	T	P	P	I
E	Q	J	M	T	I	F	Y	S
N	O	P	J	K	K	P	G	D
Z	J	S	Q	U	R	Z	O	B
J	B	Q	K	W	E	A	N	J
V	B	W	W	T	Q	R	X	I
Q	Z	M	V	T	W	N	A	N
G	A	N	Y	E	M	Q	V	O
I	O	T	A	Z	H	J	Y	S

KEY ULTIMATE

U	Q	N	O	T	F	X	K	Y
X	Z	Z	B	L	O	F	M	U
K	H	Q	R	T	I	X	Q	N
K	F	W	V	I	D	P	S	U
E	F	A	R	U	L	A	Y	Y
S	Q	L	I	F	O	H	W	I
Q	N	I	L	R	K	Z	K	Q
A	P	H	R	N	F	B	Y	K
I	J	V	E	V	T	W	M	H
T	E	B	Q	M	V	O	L	V
N	Q	V	H	P	Z	E	N	D

KEY K1 CT

D	Y	X	N	R	X	Z	C	R
F	M	R	S	V	Z	Y	Y	U
R	W	D	R	P	V	M	S	S
A	K	B	K	Z	B	C	Y	R
J	H	G	N	R	U	O	A	S
I	P	G	H	R	C	X	U	U
J	R	I	I	K	J	A	H	D
V	E	N	I	J	X	C	E	Z
P	C	X	C	R	X	G	Y	H
O	Q	X	Q	L	P	Z	P	C
K	B	A	M	G	W	L	I	J

CT MATRIX P/KEY DIAWI

E	O	F	B	I	Y	C	N	O
C	I	D	R	Q	J	Y	P	H
P	Z	D	F	W	M	C	D	T
N	I	B	F	K	P	S	L	E
V	O	D	F	E	Y	K	O	G
V	O	D	M	W	U	Z	J	M
O	C	K	D	T	E	I	E	G
D	H	T	F	F	I	N	M	E
B	N	A	R	A	W	K	P	T
I	T	E	O	B	A	X	Q	A
C	O	I	L	H	I	V	F	G

DIAWI STREAM

K	P	I	J	F	O	K	I	A
R	U	D	J	D	R	R	W	T
V	Z	Y	Q	K	T	P	P	I
E	Q	J	M	T	I	F	C	S
Z	X	D	W	S	K	P	G	D
Z	J	S	R	U	R	M	O	B
J	B	Q	Q	W	T	K	G	V
V	B	L	W	T	Q	R	X	I
Q	Z	M	V	T	W	N	A	N
G	A	N	Y	E	M	Q	V	O
I	O	T	A	Z	H	J	Y	S

U	Q	N	O	T	F	T	K	Y
M	Z	Z	B	L	O	F	M	U
K	H	Q	R	T	I	X	Q	N
K	F	W	V	I	D	P	G	U
L	O	H	B	R	L	A	Y	Y
S	Q	L	V	F	O	A	W	I
Q	N	I	E	R	N	M	Q	K
A	P	B	R	N	F	B	Y	K
I	J	V	E	V	T	W	M	H
T	E	B	Q	M	V	O	L	V
N	Q	V	H	P	Z	E	N	D

D	Y	X	N	R	X	O	C	R
S	M	R	S	V	Z	Y	Y	U
R	W	D	R	P	V	M	S	S
A	K	B	K	Z	B	C	C	R
V	A	Q	K	A	U	O	A	S
I	P	G	U	R	C	L	U	U
J	R	I	C	K	C	K	B	J
V	E	G	I	J	X	C	E	Z
P	C	X	C	R	X	G	Y	H
O	Q	X	Q	L	P	Z	P	C
K	B	A	M	G	W	L	I	J

E	O	F	B	I	Y	I	N	O
P	I	D	R	Q	J	Y	P	H
P	Z	D	F	W	M	C	D	T
N	I	B	F	K	P	S	X	E
Y	X	L	N	L	Y	K	O	G
V	O	D	Z	W	U	M	J	M
O	C	K	T	T	T	B	O	N
D	H	X	F	F	I	N	M	E
B	N	A	R	A	W	K	P	T
I	T	E	O	B	A	X	Q	A
C	O	I	L	H	I	V	F	G

Repad to 9
columns.

See previous page.

ULTIMATE mentioned in
JS interview.

Says overlay and solve
for WI(DIA) version.
WIDIA version has Row
7 as QNILRKZKQ which
has part of BERLIN
backwards. Also says
revolve, which would
place ZEND at the top left.

After conversion using
clues in appropriate
columns, KBAM
backwards is MASK.
DYXNR in first row is
DYAHR after using clues
in that matrix and the
bottom rows of the one
above it.

Note clue of scrambled
CLOCK.

Clues to use a probable
scytale transposition.

In WIDIA version, rows
4 and 5, cols. 5 thru 9,
are clues to change letter
P to letter K. See text and
matrices on page 10. This
will likely mean to change
the single letter P in matrix
S to letter K.

At this point, I want to talk about the necessity of having two streams, WIDIA and DIAWI. There were a lot of clues to change DIAWI to WIDIA and I was certain that it must be done. I was just not sure if it would occur in the original K4 position or if it would occur in the 21 shift position or in other related places in K4 and all 25 shifts of K4. Because of the necessity of having the original K4 CT as a base for determining the correct shift needed to move forward (determined in part by applying the K1 PT to related strings of text appearing alongside DIAWI in the zero shift version) I chose to apply the transposition of DIAWI to WIDIA only to the DIAWI appearing in the shift zero version of K4.

So, I was solid in my conviction that the transformation of DIAWI to WIDIA was correct and I moved forward to step F where QMEETYQME appeared vertically in the text. I remembered Ed Scheidt one time saying something about having cake meaning meet me at seven, or something like that. It reminded me of the words EAT BEEF along the first column of the K3-K4 work sheet Jim Sanborn provided for public viewing. Since MEETY ME had Qs, I knew it deserved to be looked at. It nagged at me for a while and I ultimately decided to start a stream using DIAWI unchanged in K4. When I got to step F, instead of MEET ME, I got FEED ME. That struck a bell and I decided from that point on to have two streams.

Now, Sanborn, in his notes, mentioned encountering smoke and mirrors along the way, I recognized that step F was the step he must have been referring to, as the bottom rows appeared to be a reflection of the top rows. Not a true reflection, but K on top equaled H on the bottom and H on top equaled K on the bottom, etc. At first, I couldn't figure out how to use this information, as the system of conversion of letters on the right side did not correspond with the system on the left side. I left it and moved along because it appeared to be just smoke and mirrors, as they say. Later, I went back and investigated more. I decided to do an OTP using F as CT and flipped F for key. See my file, ANKH Project, for details. The result of this was that I became convinced that both DIAWI and WIDIA streams were necessary in the scheme of things.

Well, I got up to step Q and didn't have a keyword. I remembered that Sanborn mentioned in his interview that K1 would rear its head at the end. So, just for a first try, I applied K1 CT as key to step P results. It was sort of like an OTP, but, of course, some letters of K1 were used more than once, so it wasn't a true OTP. Anyway, I got good results, but the results did not lead anywhere that I could tell.

So, I got to looking at the result of step Q and realized that there really weren't many differences in both streams. I decided to do an OTP of the WIDIA version as CT and the DIAWI version as keyword. I knew I would get mostly Ks in the resultant matrix, so I would look to see if anything stood out in what was not Ks. **Turned out that QANNQ appeared. I did the same with steps O and P. Same result. I concluded that ANN might well be a keyword. Also, in past matrices, AN and QNN strings showed with associated Qs.**

On page 6 there is a note about changing letter P in matrix S to letter K. This seems to be supported by strings such as AXQ, XQ, XP QAXP, XPK and others throughout the matrices. So, AX is turning out to be more than a keyword. Also appearing many times are strings COG, KOG, CGOG, CO, CG and others that seem to be associated with the strings mentioned above that are pointing to changing the P to a K in matrix S. Also, see pages 9, 10 and 12. As in K1 PT (modified via FIVJ changed to FIVE to read ABSENT EOF LIGHT instead of ABSENCE OF LIGHT) and the whole first half of K2 to alert analysts to insert S to obtain X, there is big emphasis now in changing letter P (the P in PRNG as hinted to by PSU in the top matrix on page 9) to letter K in matrix S.

In the meantime I looked again at step R. It seemed to give clues that pointed strongly at using a scytale to solve a transposition cipher. Three words were readily apparent: COIL in bottom left corner, GATE in bottom right corner and EOF (end of file) in upper left corner of the matrix. I looked again at step P and found ZEND on the bottom right corner and also a close approximation of REVOLVE in the text. Turning the P matrix 180 degrees would put ZEND at the top left corner. Step R had EOF in top left corner, too. Then I recalled the K3-K4 sheet that I mentioned before. It has ENDYA (YA being the last letter of Russian cyrillic) in the upper left corner, too.

1	E	N	D	Y	A	H	R	O	H	N	L	S	R	H	E	O	C	P	T	E	O	I	B	I	D	Y	S	H	N	A	I
2	A	C	H	T	N	R	E	Y	U	L	D	S	L	L	S	L	N	O	H	S	N	O	S	M	R	W	X	M	N	E	
3	T	P	R	N	G	A	T	I	H	N	R	A	R	P	E	S	L	N	N	E	L	E	B	L	P	I	I	A	C	A	E
4	W	M	T	W	N	D	I	T	E	E	N	R	A	H	C	T	E	N	E	U	D	R	E	T	N	H	A	E	O	E	T
5	F	O	L	S	E	D	T	I	W	E	N	H	A	E	I	O	Y	T	E	Y	Q	H	E	E	N	C	T	A	Y	C	R
6	E	I	F	T	B	R	S	P	A	M	H	H	E	W	E	N	A	T	A	M	A	T	E	G	Y	E	E	R	L	B	T
7	E	E	F	O	A	S	F	I	O	T	U	E	T	U	A	E	O	T	O	A	R	M	A	E	E	R	T	N	R	T	I
8	B	S	E	D	D	N	I	A	A	H	T	T	M	S	T	E	W	P	I	E	R	O	A	G	R	I	E	W	F	E	B
9	A	E	C	T	D	D	H	I	L	C	E	I	H	S	I	T	E	G	O	E	A	O	S	D	D	R	Y	D	L	O	R
10	I	T	R	K	L	M	L	E	H	A	G	T	D	H	A	R	D	P	N	E	O	H	M	G	F	M	F	E	U	H	E
11	E	C	D	M	R	I	P	F	E	I	M	E	H	N	L	S	S	T	T	R	T	V	D	O	H	W	?	O	B	K	R
12	U	O	X	O	G	H	U	L	B	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q	P	R	N	G	K	S	S	O
13	T	W	T	Q	S	I	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U	D	I	A	W	I	N	F	B	N	Y	P
14	V	T	T	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C	D	I	G	K	U	H	U	A	U	E	K	C	A	R

Watch D's + O's

TOP #4

Staple Staple Staple Staple

I then noticed the TOP#4 on the bottom. Then I saw the locations of the folds. There is a drawing in the bottom right showing how to fold the paper to display the contents at the top! Folding the paper such that the two folds are adjacent, puts the text together and this text has 99 characters! Notice that Sanborn has shifted letters so that EAT BEEF and DLOR (DOOR) are not readily seen on the sculpture. Note the code is kept secure in the two DIFFERENT SAFE DEPOSIT BOXES.

P

U	Q	N	O	T	F	X	K	Y
X	Z	Z	B	L	O	F	M	U
K	H	Q	R	T	I	X	Q	N
K	F	W	V	I	D	P	S	U
E	F	A	R	U	L	A	Y	Y
S	Q	L	I	F	O	H	W	I
Q	N	I	L	R	K	Z	K	Q
A	P	H	R	N	F	B	Y	K
I	J	V	E	V	T	W	M	H
T	E	B	Q	M	V	O	L	V
N	Q	V	H	P	Z	E	N	D

Note that when you do a 180 on this, the ZEND shows up at the upper left corner.

R

E	O	F	B	I	Y	C	N	O
C	I	D	R	Q	J	Y	P	H
P	Z	D	F	W	M	C	D	T
N	I	B	F	K	P	S	L	E
V	O	D	F	E	Y	K	O	G
V	O	D	M	W	U	Z	J	M
O	C	K	D	T	E	I	E	G
D	H	T	F	F	I	N	M	E
B	N	A	R	A	W	K	P	T
I	T	E	O	B	A	X	Q	A
C	O	I	L	H	I	V	F	G

Note the EOF in upper left corner. Also note in column 3 EAT BDDF mimicking EAT BEEF. Also, PELSK in row 4 says to change P to K.

S

E	E	N	D	Y	A	Y	S	H	N	A	I
A	A	C	H	T	N	R	W	X	M	N	E
T	T	P	R	N	G	I	I	A	C	A	E
W	M	T	W	N	H	A	E	O	E	T	
F	F	O	L	S	E	C	T	A	Y	C	R
E	E	I	F	T	B	E	E	R	L	B	T
E	E	E	F	O	A	R	T	N	R	T	I
B	B	S	E	D	D	I	E	W	F	E	B
A	E	C	T	D	R	Y	D	L	O	R	
I	T	R	K	L	M	F	E	U	H	E	
E	C	D	M	R	W	?	O	B	K	R	
U	O	X	O	G	N	G	K	S	S	O	
T	W	T	Q	S	N	F	B	N	Y	P	
V	T	T	M	Z	U	E	K	C	A	R	

Notice CEY (KEY) on both sides of O and pointing to L. This indicates to change the L in DLOR to O. Also, COYL and EOE mimic COIL and EOF in "R". Also, with a stretch, GATE is there, too.

Going to change the P to a K. See page 6 and page 10.

Folds

DOOR

This is probably the beginning of the next layer

We can see obvious relationships among all three above matrices. **The question is, can we show that the revealed 99 characters of K3 are truly derived by following all the steps that I have laid out? The answer is, YES.** If we take R and S above and compare them letter-for-letter, we can see the connection. Reverting to 11-column pads:

Legend:

- MATCH ON BOTH
- O=N, N=O SAME POS.
- R=G, SAME POS.
- E=F, SAME POS.
- A=I, I=A SAME POS.
- F=D, D=F SAME POS.
- D=I, SAME POS.

Matrix S (11-column pad):

E	N	D	Y	A	Y	S	H	N	A	I
A	C	H	T	N	R	W	X	M	N	E
T	P	R	N	G	I	I	A	C	A	E
W	M	T	W	N	H	A	E	O	E	T
F	O	L	S	E	C	T	A	Y	C	R
E	I	F	T	B	E	E	R	L	B	T
E	E	F	O	A	R	T	N	R	T	I
B	S	E	D	D	I	E	W	F	E	B
A	E	C	T	D	R	Y	D	L	O	R

WIDIA Version (Matrix R):

E	O	F	B	I	Y	C	N	O	C	I
D	R	Q	J	Y	P	H	P	Z	D	F
W	M	C	D	T	N	I	B	F	K	P
S	L	E	V	O	D	F	E	Y	K	O
G	V	O	D	M	W	U	Z	J	M	O
C	K	D	T	E	I	E	G	D	H	T
F	F	I	N	M	E	B	N	A	R	A
W	K	P	T	I	T	E	O	B	A	X
Q	A	C	O	I	L	H	I	V	F	G

So, at this point there are some questions that remain unanswered.

1. How does TUT, just now revealed by looking at keywords ANN and INSTITUTION, enter the picture?
2. How does the JOVIAN system revealed in step O affect things, if it does? I was not able to discover anything. One of the moons of Jupiter that is present in that matrix is IO. The letters IO are also part of the word IOTA. A statement by Sanborn that seems related to IOTA is to be found on page 5. At the time, I was satisfied that that was the true purpose of the O matrix and the JOVIAN system appearance there was probably meant to divert attention from IOTA. But, in retrospect, the fact that the cluster of JUPITER letters is delimited so much with Qs prompts me to want to find a significance there.
3. Is there a significance to the string FDHBVHH beyond giving credence to the fact that both the DIAWI and WIDIA streams are necessary disregarding that it reflects the ABSCISS keyword? (See file ANKH Project)
4. A couple of apparent clues show up in matrices P and Q that I have not found a use for. They may be directly related as containing pieces of keyword THEBERLINCLOCK.
 - a. In step P of the WIDIA stream, taking all of row 7, the string QNILRKZKQ is present. In the distant past, while looking at using shifts of ABSCISSA, I came across a string similarly delimited by Qs on both ends. The string was laid out backwards; I rotated the matrix 180 degrees. There was part of a word evident, but the length of the string and the nearby word EMBED, inferred to change the string to have the indicated word spelled correctly. I changed the spelling of the word in the CT and ran a substitution using the shifts of ABSCISSA. This got results showing that I did what was expected. Now, in this case of matrix P, the string QNILRKZKQ has part of the word BERLIN spelled backwards such that flipping it horizontally reveals QKZKRLINQ. In the previous-cited case, all letters, including the Qs were used to correct the spelling. In the case before us, one would conclude to include appropriate letters from the keyword THEBERLINCLOCK. So, QKZKRLINQ would become HEBERLINC. The result has not shown to be effective in any way, so far. But, it would be prudent to check further. I just looked at it again and now it appears that clues towards solving K4 are being given. The message given by the above was to rotate the matrix by 180 degrees, not to spell out THEBERLINCLOCK.
 - b. In step O of the DIAWI stream, the strings CLUU and COLKC cross at the L. Obviously, the clue is misspelled CLOCK. The right-most column of the matrix has a structure that would repeat if the rows containing O and L were to be exchanged. This process would infer to change also the rows containing the K and C below. The apparent flipping of the letters there might hint at how to set up the matrix towards solving a scytale.
5. Matrix S is apparently the final to run a transposition decryption using the scytale method, and it can be shown that the route I took through the "mask" produces a matrix that is directly related to the S matrix. But, what is the mechanism for getting there? It might not matter, but it might supply a crucial clue as to how to perform the final procedure which I have not succeeded in doing so far. I don't think this is the case, though.

So, we have potential keyword ANN (See fifth paragraph on page 7). I tried it on the S matrix using a Quagmire III substitution. The result is matrix T below.

Many clues are provided by Sanborn in his interviews. Two of his quotes are to be found below. Notice the appearances of DIE and HIT IT in both the lower quotation and in matrix T. Also note the QCL clue at the bottom and the KLU cluster at the top left and KLU again at top right. These all seem to point to the QKQ string in the third row.

I noted that Sanborn used the word INSTITUTION a few times. It didn't fit the context of the sentences and stood out as something to look at. I used it as keyword in a Quagmire III substitution also with matrix S. The result is matrix U. Notice the WAS in the bottom row. It occupies the same positions as QCL in matrix T. Also notice the QPQ in the third row. The WAS indicates the original letter P is replaced. **See side comments for matrix R on page 6.**
A change of single letter P in matrix S to letter K is indicated.

T

ENDYAY/ANN

T	K	L	U	H	C	Z	U	K	K	V
H	Y	U	E	F	B	T	L	Z	K	T
E	D	Q	K	Q	B	V	H	Y	H	M
J	Z	E	J	K	U	K	M	F	T	E
N	X	X	G	T	J	E	K	C	J	Q
M	V	O	E	I	T	M	B	D	I	E
T	M	N	X	H	B	W	K	B	W	V
I	Z	M	L	P	V	M	J	N	M	R
H	M	Y	E	L	Q	C	L	D	F	B

DECRYPT

U

ENDYAY/INSTITUTION

V	K	T	X	M	X	E	D	T	Y	V
M	J	B	K	T	W	Y	Q	P	H	M
I	D	U	I	X	E	Q	P	Q	Y	M
B	Z	X	N	T	D	F	A	J	S	E
W	F	E	Y	V	O	C	P	G	T	B
V	V	S	K	N	A	J	W	Y	P	E
V	M	S	R	M	W	C	I	F	Z	V
N	G	O	S	U	E	J	N	W	S	I
M	M	P	K	U	W	A	S	Y	K	B

DECRYPT

JS: Sure. You know, I have been trusting it to an **institution** in order to get it off-site from where I am, and so that in the event of my demise, it will leave somebody there who can say, "Yes, this is the code, you got it," or whatever. But that's not easy to do. It's not easy to find an **institution** like that.

JS: I have. But when I first started the project I said to myself: In order for me to do the right job, for this site and for this **institution**, I had to assume the same kind of role that they assume. I mean, the CIA employees keep secrets until they **die** routinely. Deep Throat not **withstanding**, **these kinds** of things can haunt you for your entire life. And I'm sure this could, but on the other hand, somebody might **hit it**. And that would be great, but **sort of game over**.

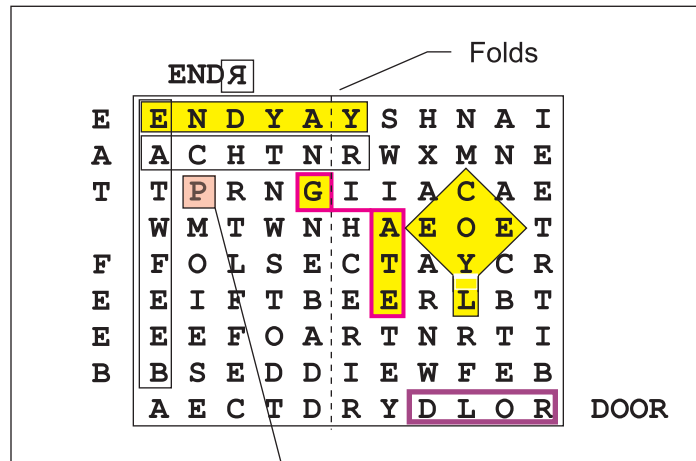
I also noticed that many of the letters in matrix T are repeated in matrix U in the same positions. I compared the keywords:

A	N	N	A	N	N	A	N	N	A	N
I	N	S	T	I	T	U	T	I	O	N

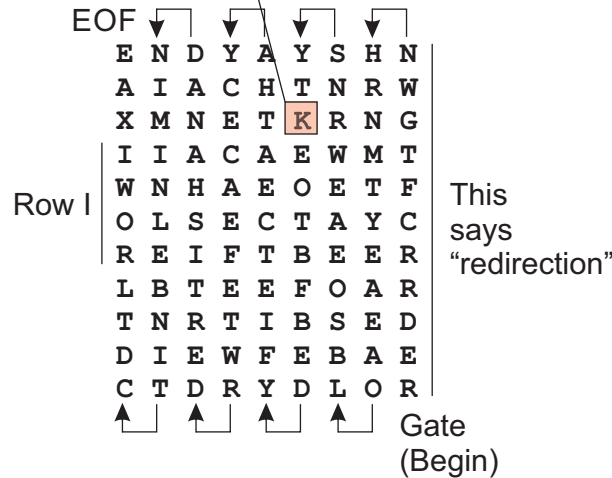
You can see that the FIRST N of ANN lines up with both Ns of INSTITUTION. Note also that in the same positions as QCL and WAS in the matrices T and U, the letters TUT appear in INSTITUTION. This cannot be simply a coincidence, since K3 talks about opening TUT's tomb. (Note: It is a coincidence only that designations of matrices T and U are related to the discovery of TUT in INSTITUTION.)

Don't forget, we have been using derivatives of the matrix S.

S



Repad to 9 columns/Creation of Scytale Strip
(Direction of choice of letters indicated by arrows)



Note: Letter P has been changed to letter K. See comment in bold face on page 11.

Reconstruction of Above Matrix
According to Apparent Instructions

Row A	R	E	D	R	R	C	F	T	G	W	N
B	H	R	N	M	T	Y	E	A	E	A	O
C	L	B	S	O	E	A	E	W	R	N	S
D	Y	T	K	E	O	T	B	F	B	E	D
E	Y	F	I	E	T	C	E	A	T	H	A
F	Y	C	E	C	A	E	F	E	T	W	R
G	D	E	R	T	I	S	H	A	N	A	D
H	N	I	M	I	N	L	E	B	N	I	T
I	C	D	T	L	R	O	W	I	X	A	E

Sanborn comment in interview:
“...**eke** that **minuscule bit**”. Confirms change from P to K and set up for scytale.

V

Using the above matrix, I was not able to come up with a good number of characters per revolution to reveal the plaintext. I also tried with the CT repadded but not redirected as above. With the exception of the change of letter P in matrix S to letter K, I believe that matrix is the one to solve.

Kryptos CT as shown
on sculpture

Letter
Count

EMUFPHZLRFAXYUSDJKZLDKRNSHGNFIVJ 32
YQTQXQBQVYUVLLTREVJYQTMKYRDMFD 31
VFPJUDEEHZWETZYVGWHKKQETGFQJNCE 31
GGWHKK?DQMC PFQZDQMMIAGPFXHQR LG 30
TIMVMZJANQLVKQEDAGDVFRPJUNGEUNA 31
QZGZLECGYUXUEENJTBJLBQCRTBJDFHRR 32
YIZETKZEMVDUFKSJHKFFWHKUWQLSZFTI 31
HHDDDUVH?DWKBFUFPWNTDFIYCUQZERE 31
EVLDKFEZMOQQJLTTUGSYQPF EUNLAVIDX 32
FLGGTEZ?FKZBSFDQVGOGIPUFXHHDRKF 31
FHQNTGPUAECNUVPDJMQCLQUMUNEDFQ 30
ELZZVRRGKFFVOEEXBDMVPNFQXEZLGRE 31
DNQFMPNZGLFLPMRJQYALMGNUVPDXVKP 31
DQUMEBEDMHDAFMJGZNUPLGEWJLLAETG 31
435

Kryptos CT showing
original matrix

Letter
Count

EMUFPHZLRFAXYUSDJKZLDKRNSHGNFIV 31
JYQTQXQBQVYUVLLTREVJYQTMKYRDMF 31
DVFPJUDEEHZWETZYVGWHKKQETGFQJNC 31
EGGWHKK?DQMC PFQZDQMMIAGPFXHQR LG 31
TIMVMZJANQLVKQEDAGDVFRPJUNGEUNA 31
QZGZLECGYUXUEENJTBJLBQCRTBJDFHR 31
RYIZETKZEMVDUFKSJHKFFWHKUWQLSZFT 31
IHDDDUVH?DWKBFUFPWNTDFIYCUQZER 31
EVLDKFEZMOQQJLTTUGSYQPF EUNLAVI 31
DXFLGGTEZ?FKZBSFDQVGOGIPUFXHHDR 31
KFFHQNTGPUAECNUVPDJMQCLQUMUNEDF 31
QELZZVRRGKFFVOEEXBDMVPNFQXEZLGR 31
EDNQFMPNZGLFLPMRJQYALMGNUVPDXVK 31
PDQUMEBEDMHDAFMJGZNUPLGEWJLLAE 31
TG 2 436

ENDYAHROHNLSRHEOCPTEOIBIDYSHNAIA 32
CHTNREYULDSL LSLNOHSNOSMRWXMNE 30
TPRNGATIHNRRARPESLNNELEBLPIIACAE 31
WMTWNDITEENRAHCTENEUDRETNHAE OET 30
TFOLSEDTIWENHAEIOYTEYQHEENCTAYCR 32
EIFTBRS PAMHHEWENATAMATEGYEERLB 30
TEEFOASFIOTUETUAEOTOARMAEERTNRTI 32
BSEDDNIAAHTTMSTEWPIEROAGRIEWFEB 31
AECTDDHIILCEIHSITEGOEAO SDDRYDLORIT33
RKLMLEHAGTDHARDPNEOHMGFMFEUHE 29
ECDMRI PFEIMEHNLSSTTRTVDOHWPOBKR 31
UOXOGHULBSOLI FBBWFLRVQQPRNGKSSO 31
TWTQSQJSSEKZZWATJKLUDIAWINFBNYP 31
VTMZFPKWGDKZXTJCDIGKUHUAEKCAR 31
434

ENDYAHROHNLSRHEOCPTEOIBIDYSHNAI
ACHTNREYULDSL LSLNOHSNOSMRWXMNE
TPRNGATIHNRRARPESLNNELEBLPIIACAE
WMTWNDITEENRAHCTENEUDRETNHAE OET
FOLSEDTIWENHAEIOYTEYQHEENCTAYCR
EIFTBRS PAMHHEWENATAMATEGYEERLB
TEEFOASFIOTUETUAEOTOARMAEERTNRTI
BSEDDNIAAHTTMSTEWPIEROAGRIEWFEB
AECTDDHIILCEIHSITEGOEAO SDDRYDLOR
ITRKLMLEHAGTDHARDPNEOHMGFMFEUHE
ECDMRI PFEIMEHNLSSTTRTVDOHW?OBKR
UOXOGHULBSOLI FBBWFLRVQQPRNGKSSO
TWTQSQJSSEKZZWATJKLUDIAWINFBNYP
VTMZFPKWGDKZXTJCDIGKUHUAEKCAR
434